The Physics of Roller Coasters from a Designer's Perspective

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Abstract

For millions of people around the world, roller coasters are a heart-pounding, adrenaline-boosting amusement park staple, providing a high level of thrill in a safe and controlled environment. Their complicated twists, turns, and drops create a feeling of euphoria that prompts riders to stand in line for hours at a time. The engineering underlying these modern marvels is highly precise, requiring not only knowledge of concepts from across physics, but an intuition about how each element of a roller coaster will impact the rider. The goal of this paper is to explore the connections between physics, engineering, and perception through a designer's perspective. This paper will investigate: basic roller coaster design considerations, G-forces and the relationship between track elements and the forces experienced by riders, the physics concepts underlying various roller coaster elements (including the electricity and magnetism concepts of LSM launch systems, the energetics of vertical loops, and the kinematics of banked turns), and lastly the effect non-physical design elements have on rider perception. In short: how do designers and engineers collaborate to create a total ride experience that is both thrilling and safe?

1 Introduction

The roller coaster is an iconic piece of engineering that is a staple attraction for any thrill-seeker. Popularized in America, they have become a central part of amusement parks everywhere. One of the earliest 'roller coasters,' shown in Fig. 1, was a repurposed railway known as the Mauch Chunk Switchback Railway of Pennsylvania. Built in 1827, cars could achieve speeds of up to 50mph when fully loaded with one and a half tons of coal. After being emptied, the cars were hauled back up to the top of the hill by mules. By the 1870s, the mines in the area had been fully excavated, ending the Mauch Chunk's time as a coal railway. Rather, it became a popular tourist destination for thousands of city-dwellers. In 1873, the Mauch Chunk carried 30,478 passengers for 75 cents apiece. Clearly, America craved thrill [29].

Figure 1: The Mauch Chunk Switchback Railway of Pennsylvania was originally used in the 19th century to carry coal, but was repurposed as a tourist attraction. It can be considered the original American roller coaster. Taken from [5].

High speeds and physical thrill are a central part of the roller coaster experience, but do not fully explain their allure. Roller coasters have always incorporated strong thematic elements as part of their appeal. LaMarcus Adna Thompson, often credited as the "Father of the American Roller Coaster," patented his *Gravity Switch Back Railway* in 1885, a slow, gravity-powered ride that coasted through a constructed scene at around 6 mph [16]. Thompson's ride has been labeled 'the first commercially successful roller coaster'; although his design was not the first to be introduced in the U.S., it is considered the beginning of the 'roller coaster frenzy' that would spread throughout America in the early 1900's. Riders of the Gravity Switch Back Railway at Coney Island were attracted not by its thrills, but rather by the opportunity to be immersed in a novel experience [13]. From these somewhat humble beginnings, the roller coaster has evolved into a machine

that can safely take people over 400 feet into the air and through a complicated series of dips, loops, and corkscrews, sometimes at speeds faster than 100mph. It takes precise computation, incredible technology, and contemporary materials to create these marvels of modern engineering.

In this paper, we will take on the roll of roller coaster designers, using a variety of physics concepts to engineer a coaster from the ground up. We will explore major elements through several different lenses, including classical mechanics, electricity and magnetism, and electronics. There will also be a brief introduction to the elements of roller coasters used for inducing psychological reactions in riders rather than physical forces.

2 Design Considerations, Safety Protocols, and Concept Drafting

The design process for a roller coaster does not begin with a long list of calculations. Rather, designers work with site owners and theme park specialists to come up with a ride concept suitable for each specific customer. The most important question we can ask as designers is: Who is the target audience and what kind of experience does the customer want them to have? For example, rides intended for children or the elderly will not be as intense as rides meant to attract teens (although thrill rides tend to be self-editing by nature, as guests develop an intuition for the level of thrill they can handle) [6]. Once the target audience is established, we can begin to draft a design concept that works with the specifications of the build site [22].

The location of a roller coaster has a great effect on its overall design. A new ride must fit into the space allotted to it and mesh with any existing rides in the park. Local building codes may designate a maximum height, or the area might experience high winds, requiring the track to be lower to the ground for safety. A ride may be completely or partially indoors (especially common with themed rides). Tracks can also utilize existing topography to cut down on building costs by moving with the land; no need to build a hill when there's one already there! In addition to local codes, roller coasters must follow a strict set of international standards and regulations specified in the ASTM International Technical Committee F24 on Amusement Rides and Devices [2]. Established in 1978, the documents establish universal safety standards, owner responsibilities, and maintenance and inspection protocols. Specifics addressed include evacuation and safety signage and detailed analysis of the interaction between the ride and the rider (such as the 'safety envelope' around the ride vehicle, which is the maximum reach of an average human[1] plus three inches in all directions from the ride vehicle) [13].

[1]Most adult rides are designed for a height range of 4' 6" to 6' 6" and can safely accommodate chest circumferences up to 52 inches. Unfortunately, some riders who fail to fit this 'average' profile and would be unable to enjoy the ride safely have to face the dreaded 'walk of shame,' sometimes after hours in line [4].

Once we have a few key points labeled, such as the maximum and minimum elevations and the track perimeter, we can begin to block in key elements, such as the station, brakes, and lift or launch sections. A rough sketch usually follows before we can transfer the design to 3D roller coaster design software that allows us to build quick 3D layouts of tracks that can then be ridden from a simulated rider's perspective. Before modern 3D design software, designers would have to draw the track layout on paper before converting their drawing to a 2D profile, which could be cut out and modeled on the site plan, much like a paper architectural model. Paper supports were connected with string and once a working model was established, measurements were taken and fed into a computer that would calculate whether or not the coaster would make it around the track. This was an iterative process; often, draft layouts would need dozens of revisions before the computer would accept the design [6]. Luckily, modeling software has drastically decreased the time needed to create a working layout that falls within the boundaries of safety specifications.

Of course, while safety is a priority for every designer, the real purpose of a roller coaster is thrill. Roller coasters offer riders the chance to engage in an activity that *feels* dangerous without actually being so. As roller coaster designer John Wardley puts it: "With a roller coaster, you can get the sensation of an exhilarating pursuit safely, quickly, cheaply, with no specialized knowledge or equipment, and it's as simple as that" [6]. Thrill rides achieve their sense of danger through both physical and psychological means. Often, these two techniques are used in conjunction to great effect. The setting and execution of a ride can have just as great an effect on the perceived thrill as the physical ride itself. This paper will primarily discuss physical thrill, since it is more dependent on the physics of the roller coaster, but to truly understand the interaction between rider and ride, we will also pay attention to psychological thrill, as it is a key part of the design process.

2.1 Physical Thrill, G-force, and Intensity

Physical thrill is the result of forces acting on the human body. The unusual combinations of forces experienced during a roller coaster ride can elicit feelings of falling, flying, or rocketing out of control. This produces adrenaline; because we perceive physical threat, the "fight-or-flight" response causes the brain to prepare the body for battle. This results in a pleasurable buzz, prompting repeat rides (and more money for the theme park). Engineering physical thrill is a delicate balancing process between the physical safety of the guests and their perceived level of excitement.

Ride elements such as drops, launches, spins, and even shaking or vibrations induce different forces that riders experience (and designers typically express) as G-forces. Roller coasters have four types of G-forces:

positive G's are the downward force riders feel (for example, at the bottom of a dip or in a banked turn), creating the feeling of heaviness[2]. For example, a positive 2G-force on a 60kg body will cause the rider to experience a weight of 120kg. Positive G-forces become uncomfortable at around +5G and may cause people to lose consciousness due to hypoxia as the heart is unable to pump sufficient blood to the brain. However, some looping coasters have brief sections of up to +6G, and fighter pilots train to withstand up to +9G [7]. Negative G's are the upward force felt as riders crest a hill, creating a feeling of lightness. The amount of time spent in negative G is nicknamed "airtime" and is highly sought after by roller coaster enthusiasts due to the novel feeling of weightlessness, which occurs at 0G. Higher magnitude negative G's occur when the train has a negative acceleration of less than -9.8 m/s^2, meaning the train would drop away from an unsecured rider, but safety restraints drag the body down faster than it would normally fall. The body is much less resilient to negative G's, and unlike with positive G's, no amount of training can increase a person's tolerance. As little as -2G can cause subconjunctival hemorrhaging (bleeding in the white of the eye); loss of consciousness usually occurs around -4G [7]. Negative G's are often the trickiest for engineers to manage since the train often does not travel at a constant speed over hills; the speed of the train will be higher when its center of gravity is lower (a higher ratio of kinetic to potential energy), so the train will slow slightly as its center rises to crest a hill. Higher speeds at the front and rear of the train translate to higher magnitude negative G's, an effect that is amplified the longer the train is.

Lateral G's are the centripetal forces felt by riders as they go around a corner, pulling them to the side as their mass attempts to continue traveling in a straight line. Unbanked turns in particular create high lateral G-forces. These are often experienced as uncomfortable for guests, so most tracks avoid too many unbanked turns.[3] The last kind of G-force, linear G's, are the forces felt by riders as the roller coaster accelerates, pushing them back into their seats as their mass resists the forward acceleration of the train.

Higher G-forces are linked with higher ride intensity, but intensity as a subjective measure is affected by a number of factors, including overall speed, height, and sequencing of elements. An individual's response to particular triggers may cause them to experience one ride as more intense than another. However, the roller coasters generally regarded as the 'most intense' usually have a standout feature that they can use as their calling card. The record for the fastest roller coaster in the world is held by *Formula Rossa* at Ferrari World, Abu Dhabi, UAE , a steel coaster meant to emulate the experience of driving a Formula 1 race car.

[2]For clarity, it's important to note that those in roller coaster design do not typically discuss *actual* forces, but rather the "forces" *experienced* by riders. Keep in mind that G-forces are not actual forces in the physical sense.

[3]The exception is a type of ride called Wild Mouse coasters, which feature a single-car roller coaster train and a track with numerous tight, unbanked turns. This is often combined with a tight layout and a number of steep drops. The resulting high lateral G-forces create the illusion that the cart is going to fly off the tracks (an effect that is often amplified by elevating the track above the ground) [6].

It utilizes a hydraulic launch system to accelerate vehicles from 0 to 149.1mph in 4.9s, which creates 1.7 linear G's on riders, who are required to wear special goggles to protect their eyes from the wind created at such high speeds. The maximum G's felt during the ride reaches 4.8G [8]. The record for highest G-forces experienced on a modern roller coaster (some antiquated coasters have been estimated to exert over 12G!) is held by *Tower of Terror* at Gold Reef City, Johannesburg, Gauteng, South Africa. At the bottom of the first drop, it exerts +6.3G on its riders [27]. *Kingda Ka* of Six Flags Great Adventure, Jackson, New Jersey, is the world's tallest operating coaster, towering at 456 feet. It is also the second-fastest coaster behind *Formula Rossa*, reaching a top speed of 128mph just after its thrilling big drop [14].

Aside from record-breaking speeds, G-forces, and drops, perceived intensity can also be affected by simply the wobble of the train or track, which can shake riders uncomfortably at high speeds. This effect is often amplified as the coaster ages and the track settles and loosens. Some roller coasters purposefully emulate wobble, intentionally creating the feeling of being out of control. This is especially common for rides that are themed after runaway vehicles (such as *Big Thunder Mountain Railroad* at Disney's Magic Kingdom, Anaheim, California) [6].

2.2 Fighting Friction

Besides G-forces, the other primary force of interest in roller coaster design is friction. Friction is defined as a non-conservative force, so the energy the train loses to friction in the form of heat and sound is not retained in the system. As a result, each successive hill or loop on a roller coaster must be shorter than all the previous hills or loops, which is a challenge we as designers must overcome in order to keep the ride engaging. While friction obviously factors into track design (as we'll see below when we analyze a vertical loop both with and without friction), most friction management occurs through maintenance of the track and trains in order to keep the ride running as intended.

The primary approach is to reduce the amount of friction in the system by lubricating the track, changing the structure and alignment of the wheel bearing, and adjusting the wheel bearing lubrication. Temperature can also affect the speed of the train; cold temperatures increase the viscosity of the grease within the wheel bearings, causing the ride to slow down [22]. Regardless of the various tricks we might use to reduce friction, energy will gradually escape the system. It remains a central challenge to maintain the thrill of the ride even as the ride vehicle gradually slows. With very few exceptions, the energy required for the cart to make it around the track is received from the initial lift hill or launch system. The tendency of a roller coaster to get slower and less exciting as the ride goes on is the antithesis of 'show business,' as designer John

Wardley puts it [6]. Therefore, we must find innovative ways of working around the inherent nature of the roller coaster, such as including sudden surprise drops as the train re-enters the station, late tight turns that induce unexpected G-forces, and saving major elements for the end of the ride. Conventionally, the loading station is the lowest point of the ride, but positioning it halfway up can allow designers to position exciting elements out of sight of the guests and shorten the height of the initial lift hill.

3 Analysis of Coaster Elements

After reviewing all of the safety specifications, assessing the site plan, and drafting a basic idea for the coaster, we can begin to analyze some of the more important elements that, when assembled, will create a thrill ride. The design team for a real coaster includes specialists from a wide range of backgrounds, including electrical, structural, and mechanical engineers, graphic designers, architects, and 'theme park specialists' with park-specific experience in a number of these areas. We can use physics to take a brief look at some of the process behind assembling a working coaster, but in reality the specialized knowledge required by each element can be vastly complex and ranges beyond the scope of this paper.

3.1 Lift Hill or Launch: Adding Energy to the System

To start the design process, we must first decide how our roller coaster is going to gain the energy it needs to speed through various elements. This occurs through two primary methods: lift hills and launches. A lift hill is any positive, powered incline designed to increase the total energy of the roller coaster by pulling the cart to a higher elevation. Traditionally, lift hills are associated with the tension-inducing clack-clack-clack noise of the anti-rollback (ARB) device on the bottom of the cart dragging over the ARB dogs on the track, devices that prevent the roller coaster train from sliding backwards down the lift hill. Modern coasters can avoid this noise, but sometimes choose to include it since it is such a potent psychological cue of the thrill to come. Interestingly, studies have shown that the highest heart rates during a roller coaster ride are often on the initial lift hill, before the riders experience any physical thrill [19]!

In contrast to the slow, apprehension-building experience of a lift hill, a launch is any system that accelerates the cart suddenly, whether on a flat surface or in an incline. According to our site plan, there's no room for a lift hill to get the coaster to a sufficient speed, so we'll choose a launch system. Some common options include cable launches, friction wheels, and, the newest, compressed air launches[4], but we will

[4]Cable launches use a small cart pulled by a cable that attaches to the underside of the train to slingshot it forward. Friction wheels are essentially parallel sets of rubber tires that grip a fin on the underside of the train to accelerate it. Compressed air

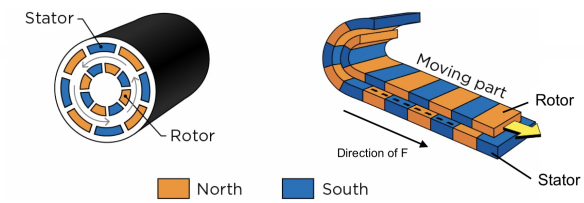

Figure 2: A Linear Synchronous Motor can be thought of as a typical electric motor that has been unrolled, so the stator (stationary) and rotor (moving) portions of the motor are side-by-side. The small arrows on the stator show the switching direction of the electromagnets, whose polarities are alternated in sequence down the track. Adapted from [20].

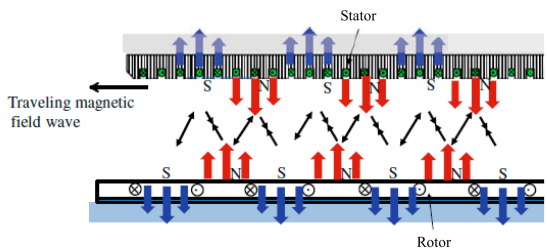

Figure 3: A close-up of the stator and rotor sections of an LSM, showing the force vectors created by the interacting fluxes of the stator and rotor. Permanent magnets of alternating polarities are attached to the rotor (the roller coaster train). The stationary magnetic fields of the rotor are first attracted then repelled as they pass by the magnetic fields created by the electromagnets in the stator. This creates a linear force described by Eq. 6.

examine a linear synchronous motor (LSM), shown in Fig. 2, launch due to its power, compressed size, and low maintenance requirements. The LSM launch utilizes the magnetic principles of attraction and repulsion to accelerate the train. Strong rare-earth magnets are attached to the underside of the train along a metal fin that sits between electromagnets on the tracks. As the train approaches the launch section, current is passed through the electromagnets in sequence to first attract, then repel the fin as it goes by, as shown in Fig. 3. These rapid switches in polarity require precise timing from a computer that controls the ride [22].

To understand the interaction between current and motor behavior, let's first consider a more traditional rotating electric motor. Electric motors operate by basic principles of electromagnetic induction; current flow through a conductor will produce a magnetic field that can be modified by varying the current magnitude or direction, described by Faraday's Law of Induction, or

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \tag{1}$$

launches again rely on a small cart to pull the train forward, but instead of being attached to a cable pulled by motors, the cart is rocketed under the track by powerful refrigerated compressed air.

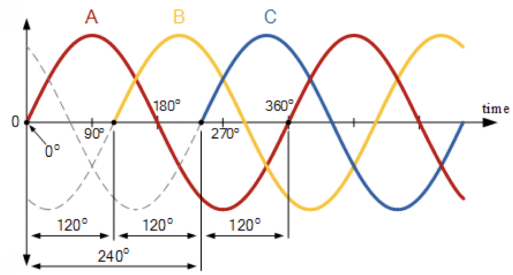

Figure 4: An example of the 3-phase waveform, where each phase is shifted in time by 120°. A waveform like the one above is used to power a 3-phase motor, pictured in Fig. 5. Taken from [24].

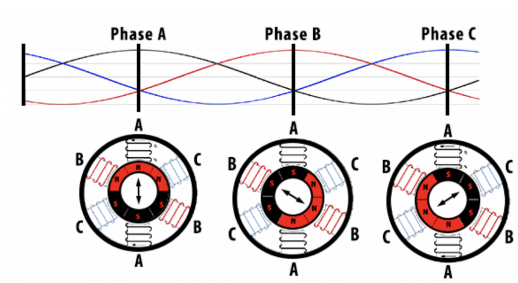

Figure 5: A traditional, rotating 3-phase electric motor, demonstrating the interaction between applied current and physical construction. Compare the 3-phase waveform to the one shown in Fig. 4 and the motor to the one pictured in Fig. 2. Taken from [9].

where $\overrightarrow{E}$ is the electric field, $\nabla \times \overrightarrow{E}$ is the curve of $\overrightarrow{E}$, and $-\frac{\partial \overrightarrow{B}}{\partial t}$ is the rate of change of the magnetic flux density vector. By passing currents that vary in a controlled way through coiled electromagnets within the electric motor, varying magnetic fields are produced. As shown in Fig. 4, a traditional 3-phase rotary electric motor is powered by 3-phase currents, each offset from the last by a time interval corresponding to 120° in phase. This 120° offset means that the polarity of each of the three electromagnets within the motor, as illustrated in Fig. 5, are never identical at any given value of θ in time. The changing magnetic fields of the stator induces a current in the rotor coils, resulting in an opposing magnetic field generated by the rotor. The rotor and stator fields attempt to align themselves, resulting in a torque applied to the motor shaft as it rotates in search of field alignment.

For every instant of time 60° apart on the graph in Fig. 4, the behavior of the generated magnetic fields, or fluxes, is dependent on the state of the 3-phase waveform powering the motor. We'll analyze how the flux changes over a full motor cycle, starting with $\theta = 0$, the origin of the graph in Fig. 4. For a waveform with

a maximum amplitude of 1, the 3 component currents at $\theta = 0°$ are

$$\begin{aligned} I_A &= 0, \\ I_B &= -\sqrt{3}/2, \\ I_C &= \sqrt{3}/2. \end{aligned} \tag{2}$$

The corresponding magnetic fluxes will be proportional to these currents. Let their values be

$$\begin{aligned} \phi_A &= 0, \\ \phi_B &= -\sqrt{3}/2\Phi_{max}, \\ \phi_C &= \sqrt{3}/2\Phi_{max}, \end{aligned} \tag{3}$$

where Φ_{max} is the maximum value of the magnetic flux due to any phase. The resultant flux, or the net flux at $\theta = 0$, is given by

$$\Psi_{\alpha\beta} = \phi_B \cos 30° + \phi_C \cos 30° = 1.5\Phi_{max}. \tag{4}$$

The phasor representation of this is shown in the inset figure. Similarly, at $\theta = 60°$, let the values of the individual fluxes be

$$\begin{aligned} \phi_A &= \sqrt{3}/2, \\ \phi_B &= -\sqrt{3}/2\Phi_{max}, \\ \phi_C &= 0, \end{aligned} \tag{5}$$

where the resultant flux has rotated 60° clockwise around the stator. Likewise, progressing another 60° through time to $\theta = 120°$ rotates the resultant flux another 60° and so on, up to a full cycle, corresponding to 6 unique active resultant flux phasors. When the motor is running, the resultant flux phasor will rotate around the stator, pulling the stationary field of the rotor forward and generating torque.

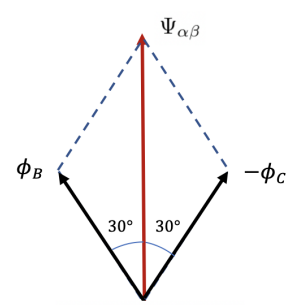

Figure 2 shows an example of a typical LSM motor, which can be thought of as an electric motor that has been unrolled so the stator and rotor are side-by-side, producing linear force along its length as opposed to torque [10]. Remember, the analysis we performed on the 3-phase motor above is precisely how an LSM works, except

instead of rotating, the resultant flux moves linearly along the length of the stator. Like other electric motors, an LSM uses electromagnets in the stator. The force produced by the motor is linearly proportional to the current and magnetic field, described by the equation

$$\overrightarrow{F} = \overrightarrow{I} L \times \overrightarrow{B}, \tag{6}$$

where $\overrightarrow{I}$ is the current, $\overrightarrow{B}$ is the magnetic field, and L is the length of the conductor. As you might imagine, accelerating a 1-2 ton vehicle to a launch speed of about 100mph in a few seconds requires a lot of force. Using Newton's second law, or

$$F = ma \tag{7}$$

to estimate, we can plug in a mass of 1 ton for m and an acceleration of about $11m/s^2$ for a, reasonable estimations for an LSM-launched coaster. This indicates that we'll need roughly 110,000N, or 25,000 pounds of force. Thus, using Eq. 6 and plugging in typical values for $\overrightarrow{I}$ L from [3] (3kA and 3m, respectively), we get a value for $\overrightarrow{B}$ in the realm of 10T![5]

Just like any other electric motor, an LSM is composed of two parts: the rotor, which generates the main magnetic field, and the stator, which generates a separate traveling magnetic field [21]. In the case of an LSM roller coaster launch, the permanent magnets on the train comprise the rotor, while the electromagnets on the track act as the stator. The stationary field of the rotor is constantly trying to catch up to the moving field of the stator, which is what 'pulls' the motor forward. The behavior of the electromagnets that generate the moving stator field is controlled by a motor component known as the output inverter. Concisely, the inverter is what controls the speed, force, and direction of an electric motor. This requires the firing of power components in the correct sequence, known as switching, in addition to some fairly complex circuitry and application-specific switching algorithms. To get a better idea of how an LSM is controlled, we can use analysis of the LSM switching algorithm from [3] as an example.

Our LSM lies on a 60m stretch of flat track directly after the loading station, low to the ground. Along the side of the track is a row of large thyristors, semiconductor devices composed of four alternating layers of p-type and n-type material that control electric current and power by acting as switches [10]. The 60m length of the launch is split into (approximately) 3m sections of wound stator connected in series. Each section comprises an electromagnet, which, when supplied with current, make up the bottom, stationary portion of the LSM (shown in Fig. 2). By dividing the 60m stator into segments, the sections not being

[5]For reference, the magnets used by the Large Hadron Collider operate at 8T, and it takes approximately 16T to levitate a frog by diamagnetic levitation of the water in its body tissues, according to the 2000 IgNobel Prize in Physics

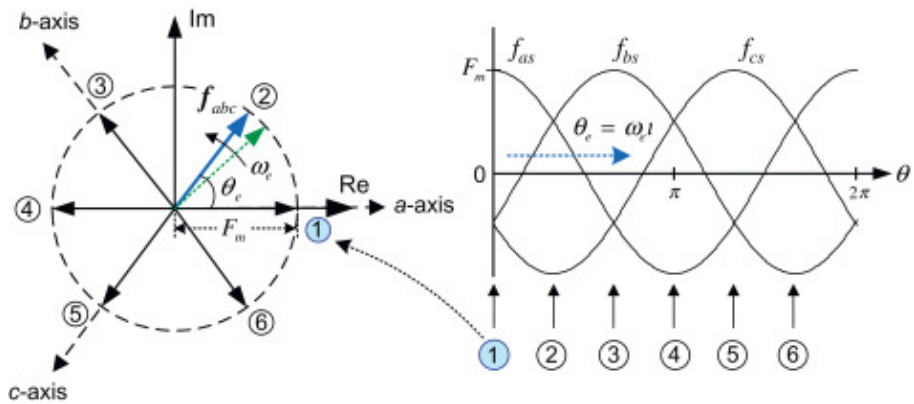

Figure 6: A diagram showing the six active voltage vectors and their corresponding locations on a 3-phase waveform. Note that there are two zero-voltage vectors located at the origin. Compare the a, b, and c axes to the phases shown in Fig. 5 and Fig. 4. f_{abc} is known as the complex space vector, or a single vector representation of the three-phase system, such as our rotating magnetic field. A space vector is different from a phasor, which is a representation of the amplitude and phase for a sine wave in the steady-state condition. f_{abc} rotates at the same angular velocity ω_e as the three phase quantities a, b, and c. Taken from [15].

utilized by the train can be shorted by the thyristors, minimizing wasted voltage [3].

In order to optimize constant power operation, the stator sections are not actually all the same length. Those towards the end of the launch, which the train passes through more quickly, are longer than those at the beginning of the launch in an effort to maintain a steady power draw. As a result, there are large asymmetric steps in inductance and resistance that make precise control of the motor current (up to 3kA!) difficult. The engineering requirements for the launch, described by [3], are as follows:

1. Smooth thrust during fast acceleration
2. Accurate motor control during the switching of the thyristors and the corresponding jumps in inductance and resistance
3. Sustained current after the maximum voltage is reached
4. Low switching frequency
5. Minimal acoustic noise

It's important to note that requirements 2 and 4 are in tension, since fast control is usually accompanied by a higher switching frequency, which allows the inverter to produce a smoother waveform that more closely resembles a sinusoidal AC signal.

In order to achieve these goals, electrical engineers must use current controllers, physical systems designed to mathematically transform the electric field to produce the desired outcome. The current control scheme is

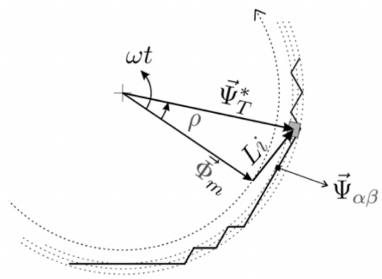

Figure 7: The flux orbit of the LSM launch. $\overrightarrow{\Psi}_T^*$ is the terminal flux, or the mathematical integral of the motor voltage. ϕ_m is the magnet flux, or the flux of the permanent magnets. $\overrightarrow{\Psi}_{\alpha\beta}$ is the resultant flux, described in eq. 4, and is the net flux in the air gap between the stator and the rotor components of the engine. In an electric motor, the terminal flux leads the magnet flux by angle ρ, or the load angle, as the current vector rotates counter-clockwise around the complex plane. An increase in load angle indicates a decrease in magnetic locking between the magnets attached to the train and the electromagnets in the track, corresponding to greater distance between our two fluxes. L_i is the inductance of the stator section. Notice that an increase in inductance also increases ρ. Taken from [3].

visualized in Fig. 6. The prime goal of the current controller is to achieve the current required by the stator while staying within the constraints of the inverter. The output of the three phase inverter corresponds to a set of $2^3 = 8$ possible states[6]. As shown in Fig. 6, six of these states are known as 'active vectors,' corresponding to a non-zero voltage magnitude, and two (located on the origin) are known as 'zero vectors,' corresponding to a voltage of magnitude 0. There are four switching rules that determine the behavior of the current controller. Its "decisions" are based on the difference between the reference current, determined by sensors that record the actual position, velocity, and acceleration of the roller coaster train, and the actual current going into the inverter. When the difference between the reference and actual current value approaches a boundary condition corresponding to a particular decision, it initiates the corresponding switch in the output voltage of the inverter.

In visual terms, the current controller is trying to draw a circle, but can only use straight lines. A perfect circle on the flux orbit diagram in Fig. 7 would correspond to a perfectly steady current over time, but would require infinite switches. Thus, the switching algorithm determines whether or not another corner is added to the flux orbit. For example, suppose as shown in Fig. 6 the state of the inverter is at $\theta = 60°$, corresponding to state 2. If the difference between the reference current and actual current indicated a stop, the inverter would switch to the nearest zero vector, the positive side of the real axis. This would cause a zero voltage on the motor terminals, meaning the mathematical integral of the motor voltage, $\overrightarrow{\Psi}_T^*$, would hold

[6]This is due to the fact that each of the 3 AC signals is either high or low for every 60° step through time, hence the binary calculation 2^3. See Fig. 6

position and no new corner would be added to the flux orbit[7]. If the difference indicated a $+60°$ switch, the inverter state would change to state 3. Since these voltage switches correspond to the direction of increase of the flux $\overrightarrow{\Psi}_T^*$, the flux orbit in Fig. 7 would get a new corner. All other active vectors shown in Fig. 6 have their own adjacent states corresponding to a $+60°$ transition, a $-60°$ transition, and a "go ahead" (the zero vector on the nearest side of the origin).

In this current control strategy, the present state of the inverter is maintained until the difference vector reaches one of the boundary values. As a result, each boundary transition involves the change of only one phase at a time. Small switches prevent the current controller from 'over-correcting', reducing switching frequency and making the system more resilient against parameter changes. This strategy also does *not* rely on a precise knowledge of motor parameters such as inductance, making it effective against the rapid asymmetric changes in inductance that occur along the length of the launch. Thus, it successfully meets the engineering requirements we laid out for our launch, allowing the LSM to smoothly accelerate thousands of passengers to breathtaking speeds!

3.2 Your Favorite Introductory Energy Problem

The next step in designing our theoretical coaster is choosing the first major element. Energy loss to friction as the ride goes on means the first track element has the most energy to 'spend,' unless we decide to add energy back into the system with an additional lift or launch midway through the ride. To use an iconic and familiar element, we'll analyze an eye-catching vertical loop and solve it in two ways: ignoring friction and then including it.

Roller coasters are a popular example in introductory physics courses to demonstrate the utility of modeling a system in terms of energy. In these idealized scenarios, the roller coaster is modeled as a closed system that neglects friction, so energy is neither added nor escapes the system in the form of heat and the transition between potential and kinetic energy is seamless. Even as we develop our model to more closely resemble real-life behavior, it is still useful to think of a roller coaster as being powered by the exchange between potential and kinetic energy. As the ride vehicle travels downhill, it exchanges some of its elevation (stored potential energy) for kinetic energy, thus accelerating.

To demonstrate the difference between an idealized and non-idealized roller coaster, let's walk through an example. Consider a simple track design with a single lift hill and circular loop, shown in Fig. 8, where

[7]It should be noted that the precise value of $\overrightarrow{\Psi}_T^*$ is unknown because the inductance of the circuit, L_i is not known to the current controller at every instant, due to the variability in stator length.

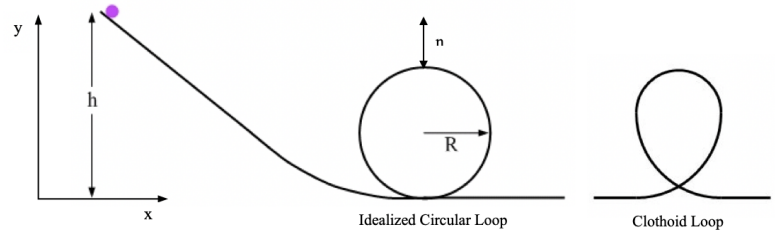

Figure 8: Our circular vertical roller coaster loop. In reality, vertical loops in roller coaster tracks are clothoid loops similar to the example on the right, shaped similarly to the center of a Venn Diagram composed of two overlapping circles. The radius of these clothoid loops is much larger at the bottom, providing a comfortable transition for riders coming out of the inversion, while the smaller radius at the top of the loop keeps riders in their seats by increasing the centrifugal force.

the bottom of the loop lies on the x-axis. We want to find the minimum initial height of the lift hill that allows the roller coaster to successfully make it through the loop. Neglecting friction, we can assume that energy is conserved, so

$$E_{initial} = E_{final}. \tag{8}$$

We also know that

$$E_{tot} = PE + KE, \tag{9}$$

where

$$PE = mgh \tag{10}$$

defines the potential energy in the system, and

$$KE = 1/2mv^2 \tag{11}$$

defines the kinetic energy. Combining Eq. 9, Eq. 10, and Eq. 11, we arrive at

$$h = \frac{v(y=0)^2}{2g}, \tag{12}$$

where m is the mass of the roller coaster train, g is the acceleration of gravity, v is the train's velocity, and h is the initial height of the train as shown in Fig. 8. While this gives us the height of the train in terms of its *velocity*, we'd really like to know it in terms of the radius of our loop, something that we already know.

From the basic concepts of circular motion, we know that the net force F_{net} will always be pointing

towards the center of the loop; if we want the train to make it all the way around the loop, we should analyze its behavior at the top, when it is upside down and F_N, the normal force the track exerts on the train, is aligned with F_g, the force due to gravity; at the bare minimum, F_N needs to be zero at this point in order for the train to stay on the track. We also know that centripetal acceleration can be defined as

$$a_{centripetal} = v^2/R. \tag{13}$$

Since F_N and F_g are aligned at the top of the loop, we can simply add them together to find the net force. Solving for the minimum net force at the top of the loop:

$$F_{net_{min}} = mg + 0 = mg, \tag{14}$$

Subbing this into $F_{net} = ma$ and rearranging some terms, we get:

$$v = \sqrt{gR}, \tag{15}$$

where v is the velocity at the top of the loop. We can then substitute this back into Eq. 12 to get:

$$\begin{aligned} x &= (\sqrt{gR})^2/2g \\ x &= R/2, \end{aligned} \tag{16}$$

where we've defined a new distance n above the top of the loop, the point we considered when we solved for velocity in Eq. 15 (as shown in Fig. 8). Adding on the height of the loop itself ($2R$), we reach the conclusion that the train must start at a height of at least $5R/2$. Since we know from Eq. 9 that the kinetic energy of the train at the bottom of the loop must be equal to its potential energy at its starting point at the top of the slope, we can rearrange Eq. 12 for $v(y = 0)$, plugging in $h = 5R/2$ to see that the speed of the train going into the loop must be at least $\sqrt{5gR}$.

3.3 The Same Loop, Now with 100% More Friction!

Although solving this problem in a frictionless realm makes for an engaging introductory problem, it makes too many simplifying assumptions for us (as roller coaster designers) to use it for real-world analysis. One of the most important omissions is friction, which quickly turns this straightforward problem into a far more difficult one. We will continue to analyze the same circular loop, but introduce sliding friction. Drawing on

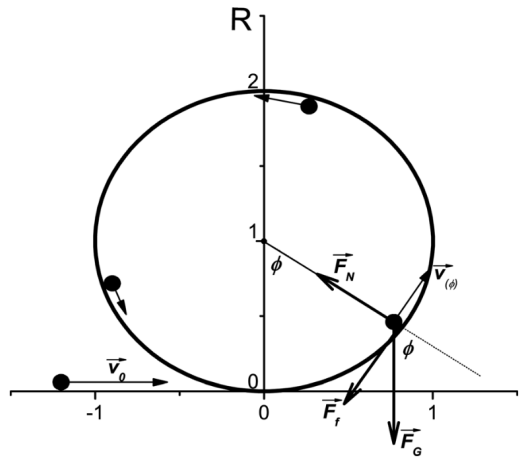

Figure 9: Our circular vertical roller coaster loop, re-parameterized for our analysis with friction. R is the radius of the loop, v_0 is the initial speed of the train leading into the loop, F_N is the normal force, F_G is the force due to gravity, F_f is the frictional force, and $v_{(\phi)}$ is the velocity of the train at a given position. Taken from [30].

analysis done by Klobus [30], we can come up with an analytical solution for the velocity leading into the loop at different values of μ, the coefficient of kinetic friction.[8]

Since the behavior of the roller coaster train at a given point is now dependent on its behavior at every previous point along the track, we must transition from the set of variables we used in our frictionless scenario to a set of parametric equations. Describing our loop from Fig. 8 in parametric form,

$$\begin{aligned} x &= R\sin\phi, \\ y &= R(1-\cos\phi), \end{aligned} \tag{17}$$

where ϕ is the train's positional angle along the loop, as shown in Fig. 9. Newton's second law $F = ma$, restated for radial motion, is

$$F_N - F_g = mv^2/R, \tag{18}$$

where F_g is the force due to gravity, m is the mass of the train, and ω is its radial velocity, and we have used the relationship $a = v^2/R$ to substitute for a. Using $F = ma$ to restate F_g in parametric form and

[8] μ depends on the properties of the two materials in contact with each other, but overall friction between the train and the track is influenced by so many factors that accounting for all possible permutations make analytical prediction and analysis very, very difficult, and something that is best left to a computer.

substituting in our variables from Eq. 17 into Eq. 18, Eq. 18 becomes:

$$F_N - mg\cos\phi = mv^2(\phi/R), \tag{19}$$

where F_g has been replaced with the vertical component of g.

Since the frictional force is described by

$$f = \mu F_N, \tag{20}$$

where μ is the coefficient of friction that incorporates the characteristics of the surface in question, we can rearrange Eq. 19 to show that

$$f = \mu(mg\cos\phi + \frac{mv^2(\phi)}{R}). \tag{21}$$

Since we are now considering friction, Eq. 8 becomes

$$KE_0 = KE(\phi) + PE(\phi) + W_{friction}(\ell(\phi)), \tag{22}$$

where KE_0 is the kinetic energy of the train at the bottom of the loop when it is at velocity v_0, $KE(\phi)$ is its kinetic energy at some position ϕ, $PE(\phi)$ is its potential energy, and $W_{friction}$ is the work done against friction at some distance $\ell(\phi)$. In integral form, work is expressed as

$$W = \int_a^b F(s)ds, \tag{23}$$

or the area under the force curve $F(s)$ between a and b. Using our variables and substituting Eq. 11 (our original definition of KE), Eq. 10 (our original definition of PE), and Eq. 23 into Eq. 22, we can say

$$\frac{mv_0^2}{2} = \frac{mv^2(\phi)}{2} + mgy(\phi) + \int_0^{\ell(\phi)} f(s)ds, \tag{24}$$

where v_0 is the speed of the train just before it enters the loop.

The real complexity of this problem arises when we consider that f at any given ϕ is dependent upon the speed of the train, which itself is dependent on f at previous points along its path. We can see this if we examine Eq. 21 and consider the change in v at particular values of ϕ over time. By expressing the integrand as a function of ϕ and arc length as

$$ds = Rd\phi, \tag{25}$$

the dependence of f on v and v on the history of the train's motion becomes more obvious. Integrating what we can of $f(s)$ from Eq. 24, we get

$$\frac{v_0^2}{2} = \frac{v^2}{2} + gR(\mu \sin\phi + 1 - \cos\phi) + \mu R \int_0^\phi v^2(\phi')d\phi'. \tag{26}$$

By introducing the factor $V = \frac{v}{\sqrt{gR}}$, we can rewrite Eq. 26 as

$$V^2(\phi) = V_0^2 + 2(\cos\phi - \mu\sin\phi - 1) - 2\mu \int_0^\phi V^2(\phi')d\phi'. \tag{27}$$

Equation 27 is a Volterra integral equation of the second kind[9], the solution of which is provided in Appendix A. Applying the solution, we find

$$V^2(\phi) = e^{-2\mu\phi}(V_0^2 + \frac{2\xi}{\zeta}) - \frac{2}{\zeta}(\xi\cos\phi + 3\mu\sin\phi), \tag{28}$$

where the substitutions $\zeta = 1 + 4\mu^2$ and $\xi = 2\mu^2 - 1$ have been made for some semblance of clarity.

We must now identify the conditions that must be met in order for the train to traverse the loop safely. We can set boundary conditions as follows:

1. For $\phi < \pi/2$, $V(\phi) > 0$
2. For $\pi/2 < \phi < 3\pi/2$, $F_N(\phi) > 0$
3. For $3\pi/2 < \phi < 2\pi$, $V(\phi) > 0$

In other words, conditions 1 and 3 dictate that the speed of the train is positive during the first and last quarters of the loop, respectively, while condition 2 ensures the train will remain in contact with the upper half of the track since the normal force F_N remains positive. In order to meet our specified conditions, we need to find values of V_0 and μ that will satisfy each condition in its given range of ϕ. This can be done analytically by identifying boundary conditions. By setting $V(\pi/2) = 0$, we obtain the boundary equation that defines a region in the μ - V_0 plane in which the train is unable to move past the first quarter of the loop due to insufficient speed. The same can be done for the final quarter by setting $V(2\pi) = 0$. This establishes the critical values of μ and V_0 where the train stops fully at the end of the track (obviously, for real engineering purposes we'd like our train to carry on a little further, but it's always helpful to start with the extreme cases).

[9]Another application for these integral equations is in ruin theory, which is the use of mathematical models to hopefully predict an insurers risk of insolvency, or ruin. Typical variables are initial surplus cash, income rate, and outgoing claims [17].

To satisfy our second condition, that the train remain on the track for the entirety of the loop, we must find an angle ϕ_0 that establishes the boundary condition $F_N(\phi_0) = 0$, meaning the train just barely makes it through the loop (an assumption that we made quite easily in our initial frictionless example). From Eq. 28, we have

$$V_0^2 > \frac{3e^{2\mu\phi}(2\mu \sin\phi - \cos\phi) - 2\xi}{\zeta}, \tag{29}$$

which means that $V_0^2\zeta$ must be greater than the maximum value of $f(\phi, \mu)$, represented by the numerator of the fraction in Eq. 29. This occurs only when $\phi = \pi$, since (returning to Eq. 21),

$$\frac{\partial f(\phi, \mu)}{\partial \phi} = 0 \tag{30}$$

only holds when $\phi = \pi$, and

$$\frac{\partial^2 f(\phi, \mu)}{\partial \phi^2}|_{\phi=\pi} < 0. \tag{31}$$

We can most easily see the results of this analysis by plotting the different outcomes on a V_0 vs. μ graph, as shown in Fig. 10. Predictably, as the coefficient of friction μ increases, the velocity V_0 must also increase in order for the train to make it through the loop. For values of μ exceeding about 0.6, the value of V_0 needed to make it through the loop rises rapidly. We can also see that in the idealized, frictionless realm where $\mu = 0$, Eq. 28 leads to the critical values of $V_0 = \sqrt{2}$ for the first quarter of the loop and $V_0 = \sqrt{5}$ for the middle half of the loop, which matches our conclusion that the initial speed of the train must be at least $\sqrt{5gR}$ as we found above!

4 Non-Physical Design Elements

4.1 Introduction to Psychological Thrill

With the physical elements of our coaster completed, we can now shift our attention to the other component of thrill: psychological thrill. Psychological thrill refers to the non-physical elements utilized to manifest a sense of fright and enhance the ride experience. Although psychological thrill may seem like the 'frosting on the cake,' more often than not non-physical elements are developed side-by-side with, or even before the physical elements of the track. Often, they are tied to the theme of the ride and are therefore sketched out in some of the earliest drafts of the roller coaster. Anyone who's watched a horror movie can attest to the power of psychological thrill; non-physical threat can be just as effective as actual peril at inducing fear.

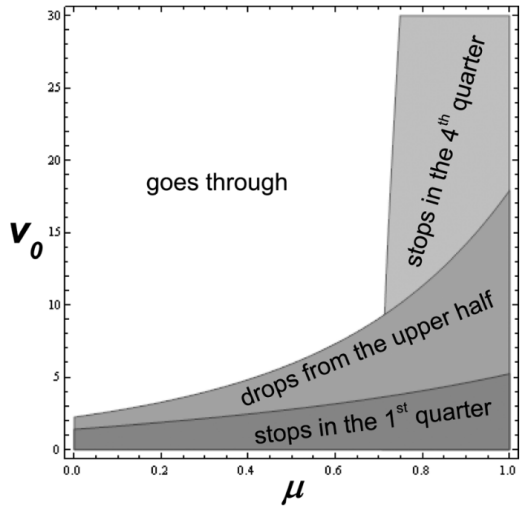

Figure 10: A graph of the four regions of the V_0-μ plane illustrating the different behaviors of the roller coaster train on a circular loop for given values of V_0 and μ. Taken from [30].

Designer John Allen explains: "You don't need a degree in engineering to design roller coasters... you need a degree in psychology" [6].

The pacing of a thrill ride can be considered the most essential form of psychological thrill. Regardless of whether we choose to theme our ride, every roller coaster tells its own sort of story through its pacing. The pacing of general thrill rides is split into two types: Constant and Punctuated. Many roller coasters are classified as Constant thrill, meaning they maintain thrill throughout the experience. Although the thrill is constant, ride elements are still interspersed with calmer sections to create contrast and give riders momentary relief. Punctuated thrill rides, on the other hand, are composed of primarily calm sections designed to build up tension to the main thrill sections (a good example is Splash Mountain, which incorporates numerous glimpses of the main drop and several short build-ups to the final big splash) [18].

Additional forms of psychological thrill include both theme elements that induce fear in the same way a horror movie might, such as jump-scares (either visual or auditory), lighting effects and/or darkness, and certain physical elements included not for the forces they create, but the perceptions they induce in the riders. Some of the most common are false-drops, small dips which lull riders into a false sense of security before the main thrill, and "headknocker" effects, or the perception that an overhang (which in reality is safely distanced above the track) appears dangerously close to fast-moving riders. A sudden dip in the track that moves the cart away from the headknocker at the final moment is particularly effective.

4.2 Augmented Dynamics

More advanced forms of psychological thrill that play directly off of rider psychology are known as augmented dynamics. Augmented dynamics describes the use of theatrical visual, tactile, and auditory cues to alter a rider's perception of how they are actually moving. One of the simplest ways to do this is through object proximity. In order to enhance the illusion of very high speeds, we can place reference points along the track. Reference points near the track will make the ride seem faster than reference points placed at a distance, since riders more accurately judge their relative speed when passing a nearby object. Reference points can be used in other, more illusory ways as well. By using projections or lights along a dark section of track, we can create the perception the ride vehicle is speeding up, slowing down, or even inverting. Cues can be tactile, as well. Intentionally adding vibration or oscillation to a track can create the illusion of a loss of control. Air cannons can be added to the start of a tunnel or the bottom of a drop to enhance feelings of speed [6].

A primary example of auditory augmented dynamics are Synchronized On-Board Audio Tracks (SOBAT). These ride-specific audio experiences have become particularly popular for the way they mesh with the physical elements of the roller coaster. In SOBAT, specialized sections of music, each with an elongated 'tail,' are stored in an onboard computer within each roller coaster train. Sensors in the track activate each section of music, with the 'tails' compensating for variations in train speed due to weight and temperature. The pacing of the music can be used to create the illusion that the ride is faster than it is. This is especially useful at the end of rides, when physical cues may be less effective because the train actually is going slower [6]. Just as in physical engineering, SOBAT uses 'safety margins' that allow audio engineers to create the best experience possible. Augmented dynamics such as SOBAT offer insight into the powerful ways our psychology impacts our experience of physical phenomena. In short, the human brain is *very* susceptible to trickery, especially when it comes to fear.

5 Conclusion

Many roller coaster designers believe we are fast approaching the limit of physical thrill. As Joe Rohde, roller coaster engineer, said, "As we move to the future, I believe that thrill rides will focus more on story because we're kind of cresting with physical sensation" [11]. Perhaps that's an indication that design and engineering should be seen as a comprehensive process that includes not just figures and specifications, but creative and artistic elements as well. Roller coaster design stands as an example of how physics, engineering

requirements, and user perception intersect to create a total experience. Undoubtedly, the ride would remain incomplete without any of these elements. It is only through integrative design practices that we have the incredible heart-pounding thrill rides we do today.

Appendix A Solution to the Volterra Integral Equation

A general form of the Volterra Intergal Equation of the second kind can be written as

$$y(x) = f(x) + \int_a^x h(t)y(t)dt,$$

the solution to which is

$$y(x) = f(x) + \int_a^x R(x,t)f(t)dt,$$

where

$$R(x,t) = h(t)^{\int_t^x h(s)ds}.$$

Taken from [1].

Appendix B The Perfect Banked Turn

While the curves in the track are not often the most memorable part of the ride, they form the connective tissue linking each of the features. Roller coaster curve design shares interesting similarities with both auto racing and flying techniques. In all three cases, the goal is to create the smoothest and most efficient ride possible, reducing strain on both equipment and passengers. For example: the best path for an auto racer to take through a high-speed turn minimizes the lateral load on the racecar by maximizing the radius through the curve. Similarly, airline pilots coordinate properly banked turns to avoid causing discomfort to passengers. The goal of the roller coaster engineer is much the same [22].

Since linear G's are usually experienced as uncomfortable by the rider, most designers attempt to minimize them by banking the turns on the track. While we might assume that simply angling the track is sufficient to reduce strain on the passengers, the axis of rotation matters a great deal if we are to design the perfect banked turn. Figure 11 shows two examples of a banked track, one outdated and one modern. In older coasters, the track was commonly rotated about its inner rail. However, this accelerates the center of mass of the passenger towards the center of the curve, which fails to eliminate the uncomfortable lateral G forces

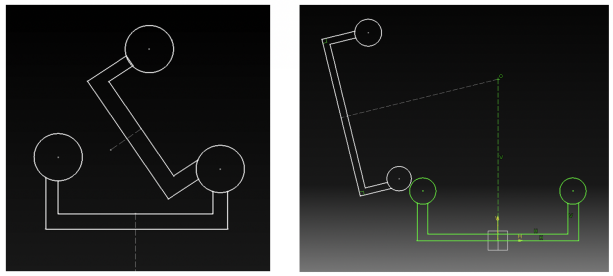

Figure 11: In early coaster designs, banking was often achieved by rotating the track around an inner rail, as shown on the left. More modern coasters rotate the track around a heartline, roughly even with riders' center of mass when seated, as shown on the right. This reduces strain on the passengers. Taken from [23].

that we are trying to avoid. To correct for this, the track should instead be rotated about the "heartline," or the point roughly above the center of the rider's torso when seated in the train. Thus, the rider's center of mass is kept at a constant position relative to the track, greatly reducing the lateral acceleration they experience [23].

In a truly ideal banked turn, one that is not overbanked nor underbanked, the roller coaster train is fully capable of staying on the track on its own, even without the force of friction. The rider will experience a a normal force pushing them to the seat of the train, as shown in Fig. 12, corresponding to some positive G force. Some curves are intentionally underbanked, either to create a thrilling effect as riders feel themselves sliding slightly to the edge of the train, or to account for a slower-than-expected ride due to lower train mass, high wind conditions, or above average friction that would turn a perfect curve into an overbanked one.

Calculating the perfect banked turn can be accomplished using kinematics. We'll imagine a scenario with no friction (truly perfect!). Per Newton's second law, we can use Eq. 13, our definition of centripetal acceleration, to say

$$F_N \sin\theta = \frac{mv^2}{r}, \tag{32}$$

where F_N is the normal force, m is the mass of the train, v is the velocity, θ is the angle of the track from perpendicular, and r is the radius of the curve. Since we are assuming no motion in the vertical direction, the sum of all vertical forces acting on the train must be zero. Therefore,

$$F_N \cos\theta = mg, \tag{33}$$

or the vertical component of F_N is equal to the train's weight. Solving Eq. 33 for F_N, substituting back

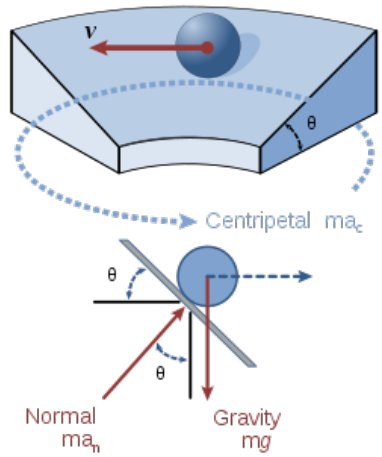

Figure 12: In a perfect banked turn, the normal force is the only force required to keep the train on the track. a_c is centripetal acceleration, a_n is acceleration perpendicular to the track, g is the acceleration of gravity, m is the mass, v is some constant speed, and θ is the angle (from perpendicular) of the bank. Taken from [23].

into Eq. 32, and eliminating m, we get

$$\tan\theta = v^2/rg, \tag{34}$$

which gives us our perfect angle! To take it a step further, we can solve Eq. 34 for v and find

$$v = \sqrt{rg\tan\theta}, \tag{35}$$

the magnitude of which is known as the 'rated speed' of the curve, or the speed at which it will remain on the curve even in the absence of friction.[10]

To put the importance of properly banked turns in context, let's pause to consider a case study. The Intimidator 305 at King's Dominion has opened in April 2010 to much excitement and fanfare. The park has invested over $25 million dollars to build the second Giga Coaster (a coaster of over 300ft tall) in America. Unfortunately, riders are soon complaining about the intensity of the first turn after the 305ft drop, the fastest on the ride. In addition, the maintenance team notices that the wheels are wearing much faster than they should, which poses a problem to the safety (and thus the future) of the ride [23].

Being savvy roller coaster designers, we know that the high speed of the first turn is not what's causing the intense forces and wear on the wheels. Instead, we suspect high angular acceleration, which translates into high vertical force. This is described by Eq.13, our definition of centripetal acceleration. Using Newton's

[10]In the real world, this is obviously a guideline and not a strict requirement, since expecting a roller coaster to maintain a constant speed through varying conditions is highly unrealistic.

second law and assuming a perfect banked turn, we can say

$$F_{vert} = ma_c = m\frac{v^2}{r}, \tag{36}$$

where F_{vert} is the vertical force on the train (and the riders). Because Eq. 36 is dependent on v and r, we have two options to reduce the vertical force: decrease the velocity of the train as it enters the turn, or increase the turn's radius.

As a short-term solution, we install a series of brakes to reduce the train's speed before it enters the turn. Although the track has been designed so the train can safely make it around at a reduced speed, the thrill of the ride is greatly reduced (especially since the turn comes right after the first major drop!). The long-term solution (and the one actually adopted in the case of the Intimidator 305) is to change the track profile. Tweaking the track after it has been built is an expensive procedure, but not necessarily unheard of. Often, the track parts have already been manufactured by the time the data has been sent to the company hired to verify the force calculations and design assumptions made [22]. Modifying the curve comes at a cost, but allows the rest of the ride to run at full speed with less guest discomfort.

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engineers, and includes numerical examples, calculations, and mathematical models of particular linear motors.

[11] J. Rhodes, T. Horn (Director), "America's Thrillmakers: Walt Disney World," Motion Picture, 2003.

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[12] J. Suul, K. Ljokelsoy, T. Midtsund, and T. Undeland, "Synchronous reference frame hysteresis current control for grid converter applications," Oct. 2010, pp. T3-111. doi: 10.1109/EPEPEMC.2010.5606801.

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[13] K. Hunt, "Design Analysis of Roller Coasters," Masters Thesis in Civil Engineering, Worchester Polytechnic Institute, May 2018.

[14] "Kingda Ka - Six Flags Great Adventure," Coasterpedia, https://coasterpedia.net/wiki/Kingda_Ka, Accessed Dec. 18, 2021.

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[16] L. A. Thompson, "Gravity Switch Back Railway," U.S. Patent No. 332,762, U.S. Patent and Trademark Office, 1885.

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[19] M. Hitti, "Roller-Coaster Ride Revs Heart Rate," WebMD. https://www.webmd.com/heart-disease/news/20070814/roller-coaster-ride-revs-heart, Accessed Oct. 27, 2021.

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[20] M. Redaelli, "Linear motors: time for some straight talking," Servotechnica, Apr. 2021, http://www.servotecnica.com/en/resources/motion-blog-en/motori-lineari-cosa-sono-e-come-funzionano/, Accessed Dec. 17, 2021.

Excellent introduction to how linear motors work and where they differ from rotary motors. Provides comparison between several types of linear motors. Includes a helpful video graphic showing a linear motor in action. Also an image citation.

[21] M. Trapanese, "A model of a linear synchronous motor based on distribution theory," Journal of Applied Physics, vol. 111, no. 7, p. 07E731, Apr. 2012, doi: 10.1063/1.3679046.

Uses distribution theory to analyze linear machines in order to account for both ideal and non-ideal influencing factors in the mathematical model. Author is probably not a native English speaker, which makes parsing some of the language difficult.

[22] N. Weisenberger, "Coasters 101: An engineer's Guide to Roller Coaster Design," Fourth Edition, United States, Nick Weisenberger, 2015.

An excellent source for basic physics (primarily mechanics) behind roller coasters and their design process. Includes history of roller coaster design as well as modern examples. Addresses the mechanics of particular elements using a 'physics for laypeople' approach.

[23] N. Weisenberger, "Coasters 101: Curves and Banking," Coaster101, Dec. 20, 2010. https://www.coaster101.com/2010/12/20/coasters101-curves-and-banking/, accessed Jan. 21, 2022.

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[24] "Rectification of a Three Phase Supply using Diodes," Basic Electronics Tutorials, Jun. 03, 2018, https://www.electronics-tutorials.ws/power/three-phase-rectification.html, Accessed Feb. 27, 2022.

Discusses the 3-phase waveform and 3-phase rectification, including circuit diagrams.

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